

Idea behind Modulation & Demodulation

✓ During Modulation,

After grouping of bits, mapper is used to map symbols to their respective phases.

What does "symbol mapping" actually mean?

Suppose the input bits are:

001

Since 8-PSK carries **3 bits per symbol**, these three bits are treated as **one symbol**.

So:

001

is not transmitted bit-by-bit.

Instead, the transmitter asks:

"Which constellation point corresponds to 001?"

From the mapping table:

Bits	Phase
000	0°
001	45°
011	90°
010	135°
110	180°
111	225°
101	270°
100	315°

we find:

001 → 45°

That's all "symbol mapping" means:

001
↓
Symbol at 45°

But how does a phase become a signal?

This is the next crucial step.

The carrier generated by the transmitter is:

$$\cos(\omega_c t)$$

Normally its phase is 0°.

Carrier:

$$\cos(\omega_c t)$$

For symbol **001**, we need phase 45°.

So the transmitter generates:

$$\cos(\omega_c t + 45^\circ)$$

instead.

This waveform is transmitted during one symbol interval.

Why do we suddenly talk about I and Q?

Because generating

$$\cos(\omega_c t + 45^\circ)$$

directly is inconvenient in hardware.

Using trigonometry:

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

Substitute:

$$A = \omega_c t$$

$$B = 45^\circ$$

We get:

$$\cos(\omega_c t + 45^\circ) = \cos 45^\circ \cos(\omega_c t) - \sin 45^\circ \sin(\omega_c t)$$

Since:

$$\cos 45^\circ = \sin 45^\circ = 0.707$$

we obtain:

$$\cos(\omega_c t + 45^\circ) = 0.707 \cos(\omega_c t) - 0.707 \sin(\omega_c t)$$

Now compare this with:

$$s(t) = I \cos(\omega_c t) - Q \sin(\omega_c t)$$

Therefore:

$$I = 0.707$$

$$Q = 0.707$$

This is where I/Q comes from

The phase 45° is represented by:

$$\text{Phase} = 45^\circ$$

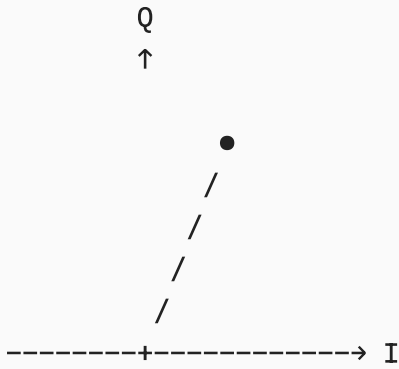
↓

$$(I, Q)$$

$$(0.707, 0.707)$$

These are simply the coordinates of the point on the unit circle.

Think about a unit circle.



The point at 45° has coordinates:

$$I = \cos(45^\circ)$$

$$Q = \sin(45^\circ)$$

which are:

$$(0.707, 0.707)$$

Another example

Suppose:

$$111$$

maps to:

$$225^\circ$$

Look at the unit circle.

225° lies in the third quadrant.

Coordinates:

$$I = \cos 225^\circ = -0.707$$

$$Q = \sin 225^\circ = -0.707$$

Thus:

111

↓

225°

↓

$(-0.707, -0.707)$

↓

$-0.707\cos(\omega t)$

+

$0.707\sin(\omega t)$

That waveform gets transmitted.

What the transmitter is really doing

Many students imagine:

Bits

↓

Phase

↓

Antenna

Actually it is:

Bits

↓

Phase

↓

I/Q coordinates

↓

$I\cos(\omega t) - Q\sin(\omega t)$

↓

RF waveform

↓

Antenna

Let's trace one complete symbol

Suppose:

```
Bits = 001
```

Symbol Mapping

```
001 → 45°
```

Convert phase to coordinates

```
45°
```

```
↓
```

```
(cos45°, sin45°)
```

```
↓
```

```
(0.707, 0.707)
```

Generate waveform

```
I=0.707
```

```
Q=0.707
```

So:

$$s(t) = 0.707 \cos(\omega_c t) - 0.707 \sin(\omega_c t)$$

Transmit

This waveform goes to the antenna.

What arrives at the receiver?

Ideally:

$$r(t) = 0.707 \cos(\omega_c t) - 0.707 \sin(\omega_c t)$$

plus some noise.

The receiver's job is now:

"Find the original I and Q."

How does the receiver find I?

Multiply by:

$$\cos(\omega_c t)$$

Receiver computes:

$$r(t) \cos(\omega_c t)$$

Substituting $r(t)$:

$$(0.707 \cos(\omega_c t) - 0.707 \sin(\omega_c t)) \cos(\omega_c t)$$

After filtering, all the high-frequency terms disappear and only:

$$0.707$$

remains.

Thus:

$$\text{Recovered I} = 0.707$$

How does the receiver find Q?

Similarly:

$$r(t) \sin(\omega_c t)$$

After filtering:

$$\text{Recovered Q} = 0.707$$

Now the receiver has:

```
(I,Q)
=
(0.707,0.707)
```

It plots that point and asks:

"Which constellation point is closest?"

Answer:

45°

Using the mapping table:

```
45°
↓
001
```

Recovered bits:

001

The single most important connection to understand is:

```
Bits
↓
Phase
↓
Point on Unit Circle
↓
(I,Q)
↓
Waveform
```


and at the receiver:

```
Waveform
↓
(I, Q)
↓
Nearest Point
↓
Phase
↓
Bits
```

Once this chain becomes clear, every PSK and QAM modulation scheme becomes much easier to understand!